

Lyapunov-like prescribed-time stability of impulsive systems via event & self-triggered impulsive control

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ABSTRACT

The main emphasis of this paper is to address the issue of event- and self-triggered impulsive control for the prescribed-time stability of impulsive systems. In the first part of the article, some Lyapunov-like sufficient conditions are derived to stabilize an impulsive system within the prescribed-time, where impulsive instants are obtained through an event-triggered mechanism. Unlike event-triggered control, event-triggered impulsive control executes solely when the event is generated. The second part of the article provides sufficient Lyapunov-like criteria for the prescribed-time stability of impulsive systems via a self-triggered mechanism. Contrary to the event-triggered impulsive control, where triggering instants are obtained through continuous or periodic monitoring of the event-triggering conditions, the proposed self-triggered impulsive control strategy can estimate the next triggering instant by the information available in the currently triggered instant. Moreover, it is demonstrated that the Zeno behavior can be excluded from both of the proposed mechanisms. Finally, two instances are provided to illustrate the theoretical results numerically and verify the veracity of the proposed methodologies.

1. Introduction

As a class of hybrid systems, impulsive systems mainly consist of three parts: an ordinary differential equation, a difference equation, and an impulse law. As a result, it serves as a connection between continuous and discrete systems. Following the pioneering work on impulsive systems and impulsive control theory reported in the books [1,2], respectively, it has received significant attention among researchers due to its wide use in various domains such as biomedical applications [3], dynamical networks [4], and secure communication [5]. There are two main types of stability problems corresponding to impulsive systems: impulsive perturbation problems and impulsive control problems. Impulsive perturbation problems mean studying the robustness of a system in the presence of destabilizing impulses. On the other hand, stabilizing a system with the help of stabilizing impulses is referred to as an impulsive control problem. The subject of impulse control has drawn much attention from scholars lately because of its high robustness, low cost, and discontinuous nature. So, control signals at certain discrete points are enough to obtain the desired outcomes. As a result, it has found extensive application in several domains, including pest management [6], relative satellite motion [7], and hyperchaos synchronization [8], etc. Hence, several eminent researchers from different fields are working on the designing of numerous impulsive control strategies for different real world models (see [9–11] and the references therein).

The notion of finite-time stability (FTS) has gained significant attention from researchers over the past few decades because of its advantages over traditional concepts such as stability, asymptotic stability, and exponential stability because of its rapid rate of convergence and ability to reject disturbances effectively. This makes it essential in various engineering applications, including

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sliding mode control [12], high-quality industrial robot control [13], and secure communication [14]. In addition to these, there are numerous scenarios, such as medical instruments, chemical plants, aerospace engineering, etc., where high precision and thus reaching the equilibrium point in some finite-time, rather than converging to some neighborhood of the origin, is required. Moreover, the term *FTS* was coined by the authors of [15], who conducted a rigorous investigation on the *FTS* of continuous autonomous systems. Since then, it has undergone extensive research and has been expanded upon in a number of models (see [16–18] and the references therein). However, the settling time in this case depends on the initial conditions, which are typically unavailable or misestimated. To address this issue, the concept of fixed-time stability ($F_\chi TS$) was introduced in [19]. The fundamental idea here is to provide a uniform upper bound for the settling time, independent of the initial conditions. Some of the intriguing findings in this aspect can be found in [20–23]. Although the settling-time is some constant depending on the parameters, it cannot be reduced arbitrarily as per the needs of the user. Hence, the concept of predefined-time stability ($P_D TS$) was introduced in the article [24]. Here, the settling-time can be reduced as per the requirements, depending on the system's tunable parameters. Some notable works on this topic can be found in [25–27] and the references therein. Though the notion of $P_D TS$ is an improved version of $FTS/F_\chi TS$, it also suffers from the drawback of the requirement of tunable parameters of the system. To work around this discrepancy, the notion of prescribed-time stability ($P_S TS$) comes into the control society through the article [28]. The center of attraction of $P_S TS$ is that the upper bound of the settling-time can be preset by the user and is independent of the initial conditions as well as the parameters. One can find some of the important articles in this regard in the articles [29–33] and the references therein. In brief, one can find the clear idea and distinctions between several notions of stability and construction of a prescribed-time stable system along with its practical applicability in the article [29]. Whereas, [30] proposed some results on local $P_S TS$ corresponding to a Lyapunov indefinite form by virtue of impulsive regulation. A walk around to the latest developments of prescribed-time control is provided in [32]. With the help of a class of regulatory functions, the authors of [33] proposed the prescribed-time synchronization of the multi-weighted and directed complex networks.

Two control strategies are commonly implemented to achieve desired system performances, namely, time-triggered and event-triggered control approaches. Regarding time-triggered control, the control signals used to be updated at random predetermined times. On the other hand, while using event-triggered control (*ETC*), the control signals are updated at the precise moments determined by some skillfully designed event-triggered mechanism (*ETM*). So, the *ETC* strategy reduces the unnecessary transmission of signals or informations and thus effectively diminishes the communication cost and computational burden. Some of the articles in this respect and the recent advances in *ETC* can be found in [34–37]. Recently, incorporating the advantages of impulsive control into *ETC*, a new control strategy named event-triggered impulsive control (*EIC*) has fascinated several researchers to work on it. Some of the indispensable works on the *EIC* are accomplished by the authors of the articles [38–42]. Specifically, in [38], Lyapunov-based sufficient conditions for *EIC* were provided for uniform stability of impulsive systems. Moreover, incorporating forced impulses with *EIC*, they also showed the global asymptotic stability of impulsive systems. Exponential stability of impulsive systems, without any involvement of forced impulses, was proposed in [39] via *EIC*. They also proposed some results regarding the maximal permissible data dropout of impulsive control under the provided event-triggered conditions. Meanwhile, [41] studied the asymptotic and exponential stability of a class of global Lipschitz non-linear systems in the presence and absence of time delay. Meanwhile, [42] provided some sufficient conditions for $P_S TS$ of switched systems under the *EIC* mechanism with the help of forced impulses. As discussed above and as per the authors' knowledge, all the articles on the *EIC* of impulsive systems focus on stability, asymptotic stability, or exponential stability. And there are no articles on *FTS* of impulsive systems via *EIC*. But, for most real-world systems, one needs to reach the origin within some finite-time intervals. Also, incorporating forced impulses into the *EIC* mechanism reduces its efficacy as it could be time-triggered impulsive control rather than being *EIC* in some scenarios. These drawbacks of the existing articles motivate us to provide some Lyapunov-based sufficient conditions for $P_S TS$ of impulsive systems by virtue of *EIC* without the involvement of any forced impulses. This constitutes the first part of the manuscript.

Though *ETC* reduces the communication cost, it suffers from the drawback of continuous or periodic monitoring of triggering conditions. For this, one needs to have specialized hardware, resulting in the consumption of huge expenses. Hence, the notion of self-triggered control (*STC*) comes to the control community. In this case, the information available at the previous triggered instant can be used to forecast the upcoming triggering instant. As a result, in the case of *STC*, neither periodic nor continuous monitoring is necessary. One can refer to the articles [43–45] to have some detailed information in this regard. Similar to the *EIC*, impulsive control with *STC* increases its efficiency and applicability. This control strategy is known as self-triggered impulsive control (*SIC*). In this aspect, [46–48] and the references therein are some of the noteworthy works. Briefly, following the *SIC* strategy, [46] proposed some sufficient conditions to achieve global ultimate boundedness and global asymptotic stability of non-linear time delay systems. Meanwhile, with the help of a comparison approach, global asymptotic stability of non-linear impulsive systems utilizing *SIC* is provided in [47]. The authors of [48] proposed some Lyapunov-based sufficient conditions to achieve global asymptotic stability of impulsive systems under the discrete-time *SIC*. However, as far as the authors' knowledge, none of the articles present in the literature have designed *SIC* for impulsive systems to attain *FTS*. But, as mentioned earlier, various engineering problems require reaching the equilibrium point in finite-time with high precision. Thus, the notions of asymptotic or exponential stability suffer in those situations to provide the desired results. Along with this, to remove the drawback of continuous or periodic monitoring in the case of *EIC*, this article has provided some Lyapunov-based sufficient conditions for $P_S TS$ of impulsive systems by virtue of *SIC*. This motivates the second part of the work.

Motivated by the aforementioned discussions, the main contributions of the present work can be outlined as follows:

1. Based on the Lyapunov method, $P_S TS$ of non-linear impulsive systems under the *EIC* strategy is studied in the manuscript. A Lyapunov-like event-triggered mechanism, and a Lyapunov-based sufficient condition for $P_S TS$ of impulsive systems, is

proposed. In contrast to the articles [38,42], which used forced impulse sequences to obtain asymptotic stability of impulsive systems and $P_S T S$ of switched systems, respectively, we have not used any forced impulse in our work. Also, different from the articles [38,39], which achieve uniform/asymptotic and exponential stability, respectively, the present work has achieved $P_S T S$ of impulsive systems. Further, the event-triggering mechanism is free from Zeno behavior.

2. To overcome the issue of continuous or periodic monitoring of the event-triggering conditions, a SIC strategy is developed to stabilize the impulsive systems within the prescribed-time. A Lyapunov-like sufficient condition, along with a new STM , is proposed. Unlike the recent work in the article [47], which established global asymptotic stability of impulsive systems via SIC , a SIC is designed to stabilize the impulsive systems within the prescribed-time. The proposed STM is also free from Zeno behavior.
3. Numerical simulations of two examples are performed to validate the efficacy of the suggested methodologies.

The rest of the article is organized as follows: Section 2 is dedicated to the introduction of some helpful notations, system descriptions, and definitions. Section 3 provides the article's primary results. Section 4 illustrates the validity of the proposed results through two examples. Section 5 concludes the article by highlighting the significant contributions and scope for future work.

2. Preliminaries

A variety of notations, system descriptions and definitions are included in this section, which serves as a preliminary reference for the rest of the article.

2.1. Notations

- $\mathbb{R}$ is the set of real numbers; $\mathbb{R}_+ = \{y \in \mathbb{R} : y > 0\}$; $\mathbb{R}_{\geq 0} = \mathbb{R}_+ \cup \{0\}$.
- $\mathbb{R}^n$ is the n -dimensional real space with the Euclidean norm $|\cdot|$.
- $\mathbb{Z}_+$ is the set of positive integers.
- $\text{sgn}(y)$ represents the signum function for $y \in \mathbb{R}$.
- For $a \in \mathbb{R}$, $[a]$ denotes the greatest integer function, i.e. $[a]$ is the nearest integer that is less than or equal to a .

2.2. System's description

Let us consider the impulsive system:

$$\begin{cases} \dot{y}(t) = f(y(t)), t \geq 0, t \neq s_k, \\ y(t) = h_k(y(t^-)), t = s_k, k \in \mathbb{Z}_+, \\ y(0) = y_0, \end{cases} \quad (1)$$

where $y \in \mathbb{R}^n$ is the state vector; $\dot{y}(t)$ is the right-hand derivative of $y(t)$; f, h_k are the continuous functions from $\mathbb{R}^n$ to $\mathbb{R}^n$ satisfying $f(0) = 0$ and $h_k(0) = 0$ for all $t \geq 0$. Further, $\{s_k, k \in \mathbb{Z}_+\}$ (in short $\{s_k\}$) is the set of impulsive instances to be determined later. It is also assumed that f and h_k fulfill all the conditions that ensure the unique existence of the solution $y(t) = y(t, 0, y_0)$ (here, 0 in $y(t, 0, y_0)$ denotes the initial time) of the system (1) in forward time (see [2]). In addition, without any loss of generality, it is assumed that the solution $y(t)$ is right continuous and has left limits at all time.

2.3. Some definitions

Definition 1 ([42]). For any $T_p > 0$, a prescribed constant, suppose the impulsive sequence $\{s_k\}$ is given. Then, the origin of system (1) is said to be prescribed-time stable ($P_S T S$) with respect to T_p , if

- for any $\epsilon > 0$, there exists $\mathcal{L}(\epsilon) > 0$ such that whenever $|y_0| < \mathcal{L}(\epsilon)$ holds, it implies $|y(t)| < \epsilon \forall t \geq 0$.
- for any $y_0 \in \mathbb{R}^n$, $y(t) \equiv 0 \forall t \geq T_p$.

Definition 2 ([25] (κ_∞ Functions)). A strictly increasing continuous function $\psi : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is said to be a κ_∞ function if $\psi(0) = 0$ and $\psi(a) \rightarrow \infty$ as $a \rightarrow \infty$. And it is denoted by $\psi \in \kappa_\infty$.

Definition 3 ([25] (κ^1 Functions)). Consider a continuous function $\psi : \mathbb{R}_{\geq 0} \rightarrow [0, 1)$ that is strictly increasing. Further, if $\psi(0) = 0$ and $\psi(r_0) \rightarrow 1$ as $r_0 \rightarrow \infty$ holds, the function ψ is called a κ^1 function, and it is then denoted by $\psi \in \kappa^1$.

3. Main results

This section of the manuscript is further divided into two subsections. The first subsection provides some Lyapunov-like sufficient conditions for $P_S T S$ of the system (1) under event-triggered impulsive control. At the same time, the second subsection consists of the establishment of some Lyapunov-like sufficient conditions for $P_S T S$ of the system (1) under the self-triggered impulsive control approach. Moreover, both the mechanism will be shown to be free from Zeno behavior.

3.1. Event-triggered based Lyapunov-like theorems

In this subsection of the manuscript, some results providing sufficient conditions P_3TS of system (1) under EIC are provided. The exclusion of Zeno behavior is also established. Let us consider the following event-triggered mechanism:

$$s_k = \inf \left\{ t \geq s_{k-1} : \psi(t) \geq L\psi(s_{k-1}) e^{-\delta_{k-1}(t-s_{k-1})} \right\}, \quad (2)$$

and the ETM stops when,

$$t > \mu T_p \text{ or } \exists s_{k-1}, s_k \text{ such that } \mu T_p - s_{k-1} \geq \epsilon_0 \text{ and } \mu T_p - s_k < \epsilon_0 \text{ holds.} \quad (3)$$

Here, $k \in \mathbb{Z}_+$ with $s_0 = 0$ and $L > 1$ is a constant real number. $T_p > 0$ is the prescribed-time and $\mu \in (0, 1)$. $\psi(t) \equiv \psi(V(y(t)))$ and $\psi(s_{k-1}) \equiv \psi(V(y(s_{k-1})))$ are the values of the κ^1 function ψ at the time t and the last event-triggered instance s_{k-1} , respectively. Moreover, the notion $\psi(t)$ or $\psi(y(t))$ is used frequently throughout the manuscript to denote $\psi(V(y(t)))$, where V is the radially unbounded Lyapunov function. And, ϵ_0 is some constant satisfying $0 < \epsilon_0 \leq \mu T_p$. Further, $\delta_{k-1} = \frac{\ln\left(\frac{LM}{\epsilon}\right)}{\mu T_p - s_{k-1}}$ denotes a positive quantity that depends on the event-triggered instance s_{k-1} and it will be derived explicitly later on (in Theorem 1). Also, it can be observed from Eq. (2) that the above-mentioned ETM runs at most for the time interval $t \in [0, \mu T_p]$, and after that, no ETM is provided. Based on these, let us derive the following results:

Remark 1. From Eqs. (2) and (3), it is clear that there is no ETM after the time $t = \mu T_p$. Further, from Eq. (3), it can also be observed that if there exists some event-triggered instance s_{k_0} such that $\mu T_p - s_{k_0-1} \geq \epsilon_0$ and $\mu T_p - s_{k_0} < \epsilon_0$ holds, then s_{k_0} is the last event-triggered instance. Also, the restriction $0 < \epsilon_0 \leq \mu T_p$ on ϵ_0 makes sure the validity of the conditions $\mu T_p - s_{k-1} \geq \epsilon_0$ and $\mu T_p - s_k < \epsilon_0$. Otherwise, if $\epsilon_0 > \mu T_p$ then, $\mu T_p - s_0 = \mu T_p > \mu T_p$, which is absurd. Moreover, the value of μ will be chosen based on the conditions of Theorem 1. Meanwhile, more insight into all the restrictions and conditions can be visualized through the following Theorems.

Remark 2. In the previous remark (Remark 1), some insight into the restrictions on the parameter ϵ_0 are discussed. However, now, let us rigorously examine the significance of the inclusion of several parameters. The parameter ϵ_0 is incorporated just to set a bound such that if the triggering instances s_{k_0-1}, s_{k_0} satisfy $\mu T_p - s_{k_0-1} \geq \epsilon_0$ and $\mu T_p - s_{k_0} < \epsilon_0$, then s_{k_0} is the last triggering instant. Otherwise, the next triggering instant, s_{k_0+1} , generated through δ_{k_0} (please refer to Eq. (2)), may be too close to μT_p , making the term δ_{k_0} tend to infinity.

Theorem 1. Assume that there exist positive constants K, a, b, p, c_k that satisfy $a < b$, $K > 1$ and $0 < p, c_k < 1$. Further let, $A = a \left\{ \frac{K-1}{K} \right\}$, $\epsilon = \left(\frac{a}{Kb} \right)^{\frac{1}{2p}}$ and $\mu_1 \in (0, 1)$ such that $(A p \mu_1 T_p)^{\frac{1}{p}} < \epsilon$ holds. Moreover, denote $M = \frac{\epsilon}{(A p \mu_1 T_p)^{\frac{1}{p}}}$ and choose $c_k = \frac{\left(\frac{\epsilon}{M} \right)}{e^{b(s_k - s_{k-1})}}$ (clearly $0 < c_k < 1$), if $\{\mu T_p - s_{k-1} \geq \epsilon_0 \text{ and } \mu T_p - s_k < \epsilon_0\}$, where $\mu = 1 - \mu_1$. Now, corresponding to the above set of parameters, if there exists κ^1 function ψ such that

- (A1) $\psi(V(h_k(y))) \equiv \psi(h_k(y)) \leq c_k \psi(V(y)) \equiv c_k \psi(y)$, $\forall y \in \mathbb{R}^n$, $k \in \mathbb{Z}_+$
and the right-upper Dini derivative of ψ along the solution $y(t) = y(t, 0, y_0)$ of system (1) satisfies
 - (A2) $D^+ \psi(V(y(t))) \equiv D^+ \psi(y(t)) \leq -a \psi^{1-p}(y(t)) + b \psi^{1+p}(y(t))$, $t \neq s_k$,
with $\{s_k, k \in \mathbb{Z}_+\}$ being the event-triggered impulsive instances generated by the ETM (2).
- Then, system (1) is P_3TS with respect to the prescribed-time T_p under the ETM (2) satisfying the condition (3).

Proof. Let us divide the non-negative real line into two parts depending on $\psi_0 \leq \epsilon = \left(\frac{a}{Kb} \right)^{\frac{1}{2p}}$ or $\psi_0 > \epsilon = \left(\frac{a}{Kb} \right)^{\frac{1}{2p}}$, where $\psi_0 = \psi(V(y_0))$ and we denote $\psi(V(y(t))) \equiv \psi(y(t)) \equiv \psi(t)$.

Case 1: $\psi_0 \leq \epsilon = \left(\frac{a}{Kb} \right)^{\frac{1}{2p}}$. In this case, we claim that $\psi(t) \leq \epsilon$ for all $t \geq 0$.

If the claim is not true, then there exists some $t > 0$ such that $\psi(t) > \epsilon$. Then denote $t^* = \inf \{t \geq 0 : \psi(t) > \epsilon\}$. Clearly, t^* cannot be an impulsive instance as $c_k \in (0, 1)$ for every $k \in \mathbb{Z}_+$. Hence one can obtain that $\psi(t^*) = \epsilon$ and $\psi(t) \leq \epsilon$ for all $t \in [0, t^*]$, and $D^+ \psi(t)|_{t=t^*} \geq 0$. But, from condition (A2), one can derive that

$$\begin{aligned} D^+ \psi(t)|_{t=t^*} &\leq -a \psi^{1-p}(t^*) + b \psi^{1+p}(t^*) \\ &= a \psi^{1-p}(t^*) \left\{ -1 + \frac{b}{a} \psi^{2p}(t^*) \right\} \\ &= -a \left\{ \frac{K-1}{K} \right\} \psi^{1-p}(t^*) \\ &< 0, \end{aligned}$$

which is a contradiction. Hence, the claim is true, and thus the region $Y = \{y \in \mathbb{R}^n : \psi(t) \leq \epsilon\}$ is a stable region. Further, from condition (A2), it can be deduced that

$$D^+ \psi(t) \leq -a \psi^{1-p}(t) + b \psi^{1+p}(t)$$

$$\begin{aligned}
&= a\psi^{1-p}(t) \left\{ -1 + \frac{b}{a}\psi^{2p}(t) \right\} \\
&\leq -A\psi^{1-p}(t),
\end{aligned} \tag{4}$$

where $A = a \left\{ \frac{K-1}{K} \right\}$.

Case 2: $\psi_0 > \epsilon$. From condition (A2), one can deduce that

$$D^+\psi(t) \leq b\psi(t), t \neq s_k. \tag{5}$$

Since the prescribed-time T_p and the constants A, p, ϵ are known, we can select $\mu_1 \in (0, 1)$ such that

$$(Ap\mu_1 T_p)^{\frac{1}{p}} < \epsilon.$$

Denote $M = \frac{\epsilon}{(Ap\mu_1 T_p)^{\frac{1}{p}}}$ and choose $\mu = 1 - \mu_1$. As $M > 1$, $\frac{\epsilon}{M} < \epsilon$. Now, two sub-cases may arrive depending on the existence of event instances s_{k_0-1}, s_{k_0} such that $\mu T_p - s_{k_0-1} \geq \epsilon_0$ and $\mu T_p - s_{k_0} < \epsilon_0$ or for all event instances $\mu T_p - s_{k-1} \geq \epsilon_0$.

Sub-case 1: There exists event instances s_{k_0-1}, s_{k_0} such that $\mu T_p - s_{k_0-1} \geq \epsilon_0$ and $\mu T_p - s_{k_0} < \epsilon_0$ holds. Then, by virtue of Eq. (5), one can have that

$$\psi(t) \leq \psi_{k_0-1} e^{b(t-s_{k_0-1})}, t \in [s_{k_0-1}, s_{k_0}). \tag{6}$$

Using condition (A1), we have

$$\psi_{k_0} \leq c_{k_0} \psi(s_{k_0}^-),$$

and using Eq. (6), one can deduce that

$$\psi_{k_0} \leq c_{k_0} \psi_{k_0-1} e^{b(s_{k_0}-s_{k_0-1})}. \tag{7}$$

But we have, $c_{k_0} = \frac{(\frac{\epsilon}{M})}{e^{b(s_{k_0}-s_{k_0-1})}}$ and $\psi(s_{k_0-1}) \leq 1$. And thus combining Eqs. (6) and (7), it can be deduced that

$$\psi_{k_0} \leq \frac{(\frac{\epsilon}{M})}{e^{b(s_{k_0}-s_{k_0-1})}} e^{b(s_{k_0}-s_{k_0-1})} = \frac{\epsilon}{M}. \tag{8}$$

From Eq. (4), $D^+\psi(t)$ is not positive definite in the region $Y = \{y \in \mathbb{R}^n : \psi(t) \leq \epsilon\}$. But, $\frac{\epsilon}{M} < \epsilon$ and so, $D^+\psi(t)$ is also not positive definite in the region $Y_0 = \{y \in \mathbb{R}^n : \psi(t) \leq \frac{\epsilon}{M}\}$. Thus, by virtue of Eq. (8), we can obtain that $\psi(\mu T_p) \leq \frac{\epsilon}{M}$. So, s_{k_0} is the last triggering instance, and there is no ETM after the time $t = s_{k_0}$. Further, let us define δ_{k-1} as $\delta_{k-1} = \frac{\ln(\frac{LM}{\epsilon})}{\mu T_p - s_{k-1}}$ for $k = 1, 2, \dots, k_0$. Note that, in the proof of *Sub-case 1*, the value of δ_{k-1} is not used explicitly. However, to obtain the triggering instants s_k for $k = 1, 2, \dots, k_0$ one needs to define the ETM (2), and thus the value of δ_{k-1} .

Sub-case 2: $\mu T_p - s_k \geq \epsilon_0$ for every triggering instant s_k . And thus, the ETM (2) runs up to $t \leq \mu T_p$. Here also, let us define δ_{k-1} as $\delta_{k-1} = \frac{\ln(\frac{LM}{\epsilon})}{\mu T_p - s_{k-1}}$. Meanwhile, let $s_{k'}$ be the last event-triggered instant in this case. Then, from ETM (2), it can be obtained that

$$\psi_{k'} = L\psi(s_{k'-1}) e^{-\delta_{k'-1}(s_{k'}-s_{k'-1})},$$

and at $t = \mu T_p$,

$$\psi(\mu T_p) \leq L\psi(s_{k'}) e^{-\delta_{k'}(\mu T_p - s_{k'})}. \tag{9}$$

Using $\delta_{k'} = \frac{\ln(\frac{LM}{\epsilon})}{\mu T_p - s_{k'}}$ and $\psi(s_{k'}) \leq 1$, in Eq. (9) we can obtain

$$\begin{aligned}
\psi(\mu T_p) &\leq L e^{-\frac{\ln(\frac{LM}{\epsilon})}{\mu T_p - s_{k'}}(\mu T_p - s_{k'})} \\
&= \frac{\epsilon}{M}.
\end{aligned}$$

So, in both the sub-cases 1 and 2, it is obtained that $\psi(\mu T_p) \leq \frac{\epsilon}{M}$, i.e., the stable region $Y_0 = \{y \in \mathbb{R}^n : \psi(t) \leq \frac{\epsilon}{M}\}$ is reached within the time $t = \mu T_p$.

Meanwhile, since $\frac{\epsilon}{M} < \epsilon$, one can find from Eq. (4) that

$$D^+\psi(t) \leq -A\psi^{1-p}(t),$$

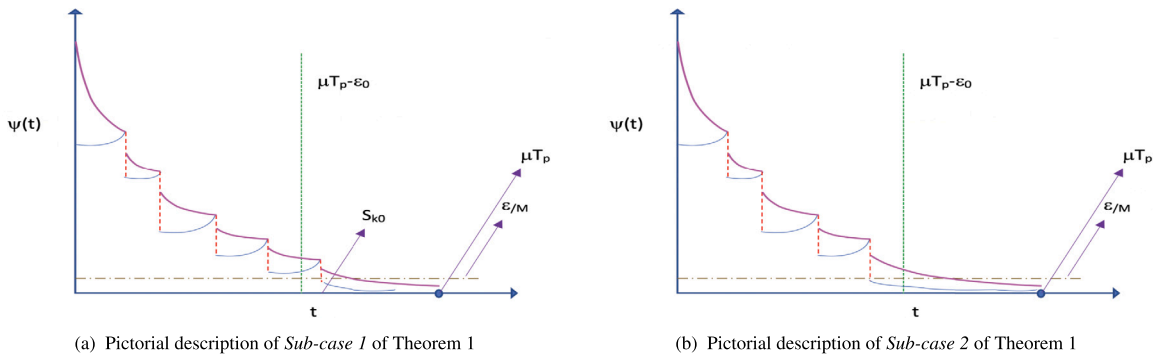


Fig. 1. Graphical illustration of Remark 3.

which on integration gives

$$\psi^p(t) \leq \left(\frac{\epsilon}{M}\right)^p - Ap(t - \mu T_p).$$

Hence, at time $t = T_p$,

$$\psi^p(T_p) \leq \left(\frac{\epsilon}{M}\right)^p - ApT_p(1 - \mu). \quad (10)$$

But we have, $\frac{\epsilon}{M} = (Ap\mu_1 T_p)^{\frac{1}{p}}$ and $1 - \mu = \mu_1$, which reduces the Eq. (10) to

$$\psi^p(T_p) \leq 0.$$

But, $\psi^p(t)$ cannot be negative for any $t \geq 0$. Hence, $\psi(T_p) = 0$. So, starting from outside the stable region, $\psi_0 \leq \epsilon$, one can obtain $\psi(T_p) = 0$ within the prescribed-time T_p and stay there afterward. Therefore, it is obvious that if we start from inside the stable region $\psi_0 \leq \epsilon$, we will obtain $\psi(T_p) = 0$ within the prescribed-time T_p . Thus, in either case, we have $\psi(T_p) = 0$. Moreover, by the definition of κ^1 functions (definition 3), we have

$$y(T_p) = 0 \text{ and } y(t) = 0 \forall t \geq T_p.$$

This completes the proof. $\square$

Remark 3. The basic idea behind the construction of such an ETM (Eqs. (2) & (3)) is to maneuver the exponential rate of convergence δ_{k-1} in such a way that the upper bound of the function $\psi(t)$, which is $L\psi(s_{k-1})e^{-\delta_{k-1}(t-s_{k-1})}$, is less than or equal to $\frac{\epsilon}{M}$ at the time of instance $t = \mu T_p$. Moreover, as described in sub-cases 1 and 2 of Theorem 1, there can be two ways of obtaining $\psi(\mu T_p) \leq \frac{\epsilon}{M}$ depending on the difference between the impulsive instants, obtained through the ETM (2) and the time $t = \mu T_p$. Also, at each impulsive instance s_{k-1} , $\psi(s_{k-1}) < L\psi(s_{k-1})e^{-\delta_{k-1}(s_{k-1}-s_{k-1})} = L\psi(s_{k-1})$. Some more insight into this can be found in Fig. 1.

Remark 4. From Theorem 1, it is clear that μT_p is the maximum time within which the trajectories should reach the stable region $Y_0 = \{y \in \mathbb{R}^n : \psi(t) \leq \frac{\epsilon}{M}\}$. Now, increasing μ means decreasing μ_1 and thus decreasing $\frac{\epsilon}{M}$ (as $\frac{\epsilon}{M} = (Ap\mu_1 T_p)^{\frac{1}{p}}$). In other words, we are giving ourselves more time to reach some closer neighborhood of the origin. Further, there is one more factor that should be considered, and it is the rate of convergence, $\delta_{k-1} = \frac{\ln(\frac{LM}{\epsilon})}{\mu T_p - s_{k-1}}$, of the ETM (2). It is also easy to observe that δ_{k-1} increases with an increase in μ . Considering all these factors into account, it can be ensured that increasing the value of μ will cause more triggering. Now let us focus on the effects of the parameter ϵ_0 on the ETM (2). The necessity of the introduction of the parameter has been elaborated in Remark 2. Also, the impulsive strength, c_k , is dependent on the value of the parameter ϵ_0 (please refer to Theorem 1). It can be seen from there that increasing the value of ϵ_0 will reduce the number of triggering instances and thus need stronger impulsive strength (i.e. a smaller value of c_k) to achieve our goal. For example, if one chooses $\epsilon_0 = \mu T_p$, then $k = 1$, and thus we will reach the region $Y_0 = \{y \in \mathbb{R}^n : \psi(t) \leq \frac{\epsilon}{M}\}$ after the first triggering instant. But, this will require a strong impulse. This is elaborated as follows:

We have $\epsilon_0 = \mu T_p = (1 - \mu_1) T_p$, which gives $\mu_1 = \frac{T_p - \epsilon_0}{T_p}$. But we have, $c_k = \frac{(\frac{\epsilon}{M})}{e^{b(s_k - s_{k-1})}} = \frac{(Ap\mu_1 T_p)^{\frac{1}{p}}}{e^{b(s_k - s_{k-1})}} = \frac{(Ap(T_p - \epsilon_0))^{\frac{1}{p}}}{e^{b(s_k - s_{k-1})}}$. So, increasing the value of ϵ_0 will require stronger impulsive control to achieve the goal. Hence, it is better to choose the value of ϵ_0 as small as one can.

Theorem 2. The system (1) is free from the Zeno behavior under the ETM (2) with (3) corresponding to the set of parameters and conditions defined in Theorem 1.

Proof. Without any loss of generality, let us denote $\psi(V(y(t))) \equiv \psi(y(t)) \equiv \psi(t)$ and $\psi(V(y(s_k))) \equiv \psi(y(s_k)) \equiv \psi_k$. Now, from the condition (A2) and the property of κ^1 function ψ , one clearly has

$$D^+\psi(y(t)) \leq b\psi^{1+p}(y(t)) \leq b\psi(y(t)), t \neq s_k. \quad (11)$$

So, for $t \in [0, s_1)$, one can obtain that

$$\psi(t) \leq \psi(y(0))e^{bt} \equiv \psi_0 e^{bt}.$$

Similarly, for $t \in [s_{k-1}, s_k)$, it can be obtained that

$$\psi(t) \leq \psi_{k-1} e^{b(t-s_{k-1})}. \quad (12)$$

Hence, combining Eqs. (2) and (12), one can acquire

$$L\psi_{k-1} e^{-\delta_{k-1}(s_k-s_{k-1})} \leq \psi_{k-1} e^{b(s_k-s_{k-1})},$$

which can further be generalized as

$$s_k - s_{k-1} \geq \frac{\ln(L)}{\delta_{k-1} + b}. \quad (13)$$

Now from, sub-case 1 of Theorem 1, we have $\delta_{k-1} = \frac{\ln(\frac{LM}{\epsilon})}{\mu T_p - s_{k-1}}$ for $k = 1, 2, \dots, k_0$ and there is no ETM after $t = s_{k_0}$. So, there is no existence of the term $\delta_{k_0} = \frac{\ln(\frac{LM}{\epsilon})}{\mu T_p - s_{k_0}}$, for which the denominator may have come closer to 0, making the term δ_{k_0} to infinite, and would have contradicted the statement of Theorem 1. So, in this case, we have $\delta_{k-1} = \frac{\ln(\frac{LM}{\epsilon})}{\mu T_p - s_{k-1}}$ for $k = 1, 2, \dots, k_0$. But, $\mu T_p - s_{k-1} \geq \epsilon_0$ for $k = 1, 2, \dots, k_0$ which implies,

$$\delta_{k-1} \leq \frac{\ln(\frac{LM}{\epsilon})}{\epsilon_0}, \dots, \text{for } k = 1, 2, \dots, k_0. \quad (14)$$

Combining Eq. (13) with Eq. (14), one can have

$$s_k - s_{k-1} \geq \frac{\ln(L)}{\frac{\ln(\frac{LM}{\epsilon})}{\epsilon_0} + b}, \dots, \text{for } k = 1, 2, \dots, k_0. \quad (15)$$

And from sub-case 2 of Theorem 1, one can have $\mu T_p - s_k \geq \epsilon_0$ for every triggering instant s_k and $\delta_{k-1} = \frac{\ln(\frac{LM}{\epsilon})}{\mu T_p - s_{k-1}}$. Which, in combination with Eq. (13) gives,

$$s_k - s_{k-1} \geq \frac{\ln(L)}{\frac{\ln(\frac{LM}{\epsilon})}{\epsilon_0} + b}, \quad (16)$$

for every triggering instant s_k .

Hence, from Eqs. (15) and (16), one can conclude that there is no Zeno behavior corresponding to the ETM (2) with (3). $\square$

Remark 5. As mentioned earlier, authors of [38] have provided an ETM to study uniform stability, whereas a forced impulse time sequence was incorporated with the proposed ETM to achieve global asymptotic stability. And, [40] proposed a dynamic event triggering mechanism to achieve asymptotic stability of unstable non-linear systems via impulsive control with the help of forced impulses. Further, an ETM incorporated with forced impulse sequence has been proposed in [42] to obtain the prescribed-time stability of switched systems. Though all of the articles mentioned above have contributed significantly to the field of control theory, the use of forced impulses somehow diminishes their efficiency, as the impulsive sequence may be generated entirely by the forced impulses. In that case, it becomes equivalent to the time-triggered mechanism. Meanwhile, an ETM to achieve exponential stability has been proposed in [39] with the help of an exponential convergent function. Whereas, [41] designed an EIC strategy to obtain asymptotic and exponential stability of a class of global Lipschitz non-linear systems in the presence or absence of delay.

However, in the present work, as given above, we have proposed an ETM where the main advantage is that it can achieve $P_S T S$ of the impulsive systems (1) without any involvement of forced impulse instants.

3.2. Self-triggered based Lyapunov-like theorems

This subsection of the manuscript consists of some results providing Lyapunov-like sufficient conditions for $P_S T S$ of the system (1) under SIC . The exclusion of Zeno behavior is also established. Let us consider the following self-triggered mechanism:

$$s_k = s_{k-1} + \frac{1}{L_0 b} \int_{\psi_{k-1}}^{\frac{M}{\epsilon} \psi_{k-1}} \frac{d\psi}{\psi} \quad (17)$$

and let it run as long as

$$s_k \leq \mu T_p. \quad (18)$$

Where, $\{s_k\}$ is the impulse instances obtained from the above STM with $s_0 = 0$; ϵ, M, μ, T_p are constants as defined in the previous Section 3.1 and $\psi_{k-1} \equiv \psi(V(y(s_{k-1})))$. And, $L_0 \geq 1$ is a constant such that $\frac{1}{L_0 b} \ln\left(\frac{M}{\epsilon}\right) \leq \mu T_p$.

Theorem 3. *There is no Zeno behavior under the STM (17).*

Proof. Eq. (17) gives

$$\begin{aligned} s_k - s_{k-1} &= \frac{1}{L_0 b} \ln\left(\frac{\frac{M}{\epsilon} \psi_{k-1}}{\psi_{k-1}}\right) \\ &= \frac{1}{L_0 b} \ln\left(\frac{M}{\epsilon}\right) \text{ for any } k \in \mathbb{Z}_+. \end{aligned}$$

Hence, the difference between two impulse instances is $\frac{1}{L_0 b} \ln\left(\frac{M}{\epsilon}\right)$, which is greater than zero. Thus, STM (17) is free from Zeno behavior. $\square$

Remark 6. As we have $\frac{1}{L_0 b} \ln\left(\frac{M}{\epsilon}\right) \leq \mu T_p$, there exists at least one impulsive instance within the interval $(0, \mu T_p]$. Also, the number of impulsive instances in the interval $(0, \mu T_p]$ depends upon A, p, μ_1, T_p, L_0, b , i.e., it depends upon all the parameters used throughout the discussion of the article.

Theorem 4. *Let there exist constants $\epsilon, M, \mu, T_p, p, A, a, b$ such that the condition (A2) of Section 3.1 holds. Further, suppose that there*

exists the constant $c \leq e^{\lfloor \frac{\ln(\frac{M}{\epsilon})}{L_0 b \mu T_p} \rfloor \{ \ln(\frac{\epsilon}{M}) - b \mu T_p \}}$ (clearly $c \in (0, 1)$ as $\lfloor \frac{\ln(\frac{M}{\epsilon})}{L_0 b \mu T_p} \rfloor > 0$) and κ^1 function ψ such that

$$(B1) \quad \psi(V(h_k(y))) \equiv \psi(h_k(y)) \leq c\psi(V(y)) \equiv c\psi(y), \quad \forall y \in \mathbb{R}^n, \quad k \in \mathbb{Z}_+,$$

with $\{s_k, k \in \mathbb{Z}_+\}$ being the self-triggered impulsive instances generated by the STM (17) is satisfied.

Then, the system (1) is $P_S T S$ with respect to the prescribed-time T_p under the STM (17).

Proof. Without any loss of generality, let us denote $\psi(y(t)) \equiv \psi(t)$ and $\psi(y(s_k)) \equiv \psi_k$. Now, by virtue of condition (A2) and property of κ^1 function, one can easily have that,

$$D^+ \psi(t) \leq b\psi^{1+p}(t) \leq b\psi(t), \quad t \neq s_k. \quad (19)$$

Since, $s_0 = 0$, therefore, from Theorem 3, one definitely has $s_1 = \frac{1}{L_0 b} \ln\left(\frac{M}{\epsilon}\right)$. Now let there be N triggering instances generated through the STM (17) and s_N be the last impulse instant. Depending on the positioning of the s_N , two cases arise, either $s_N = \mu T_p$ or $s_N < \mu T_p$.

Case i: $s_N = \mu T_p$.

In this case, for $t \in [s_{N-1}, \mu T_p)$, one can derive from Eq. (19) that

$$\begin{aligned} \psi(t) &\leq \psi_{N-1} e^{b(t-s_{N-1})} \\ &= c^{N-1} \psi_0 e^{bt} \\ &\leq c^{N-1} e^{bt}. \end{aligned}$$

And thus, by condition (B1), we have at $t = \mu T_p$,

$$\begin{aligned} \psi(\mu T_p) &\leq c\psi(\mu T_p^-) \\ &= c^N e^{b\mu T_p}. \end{aligned} \quad (20)$$

Case ii: $s_N < \mu T_p$.

For $t \in [s_N, \mu T_p] \equiv [s_N, \mu T_p]$ (as μT_p is not an impulsive instance), one can have from Eq. (19) that,

$$\psi(t) \leq \psi_N e^{b(t-s_N)}$$

$$\begin{aligned}
&= c^N \psi_0 e^{bt} \\
&\leq c^N e^{bt}.
\end{aligned}$$

So, at $t = \mu T_p$,

$$\psi(\mu T_p) \leq c^N e^{b\mu T_p}. \quad (21)$$

From Eqs. (20) and (21), it can be concluded that in both cases,

$$\psi(\mu T_p) \leq c^N e^{b\mu T_p}. \quad (22)$$

Further, using Theorem 3, we have the number of impulsive instances in the interval $(0, \mu T_p]$ is given by

$$N = \left\lfloor \frac{L_0 b \mu T_p}{\ln\left(\frac{M}{\epsilon}\right)} \right\rfloor. \quad (23)$$

Also, by Remark 6 and Eq. (23) $N \geq 1$. The condition $N \geq 1$ is essential as $N = 0$ will imply no triggering instant, and thus no impulse controller will be activated within the time $(0, \mu T_p]$. Thus, we may fail to achieve our goal within the time μT_p . Now, using

Eq. (23) and value of $c = e^{\left\lfloor \frac{L_0 b \mu T_p}{\ln\left(\frac{M}{\epsilon}\right)} \right\rfloor \left\{ \ln\left(\frac{\epsilon}{M}\right) - b\mu T_p \right\}}$ in Eq. (22), one can obtain

$$\begin{aligned}
\psi(\mu T_p) &\leq \left\{ e^{\left\lfloor \frac{L_0 b \mu T_p}{\ln\left(\frac{M}{\epsilon}\right)} \right\rfloor \left\{ \ln\left(\frac{\epsilon}{M}\right) - b\mu T_p \right\}} \right\}^{\left\lfloor \frac{L_0 b \mu T_p}{\ln\left(\frac{M}{\epsilon}\right)} \right\rfloor} e^{b\mu T_p} \\
&= e^{\left\{ \left\lfloor \frac{L_0 b \mu T_p}{\ln\left(\frac{M}{\epsilon}\right)} \right\rfloor \left\{ \ln\left(\frac{\epsilon}{M}\right) - b\mu T_p \right\} + b\mu T_p \right\}} \\
&= \frac{\epsilon}{M}.
\end{aligned} \quad (24)$$

But, as mentioned in Theorem 1, $Y_0 = \left\{ y \in \mathbb{R}^n : \psi(t) \leq \frac{\epsilon}{M} \right\}$ is a stable region and from Eq. (24) we obtained that we have reached the stable region Y_0 within the prescribed-time μT_p . Once one reaches the region $Y_0 = \left\{ y \in \mathbb{R}^n : \psi(t) \leq \frac{\epsilon}{M} \right\}$, a similar procedure to Case 1 of Theorem 1 can be followed to make sure $y(T_p) = 0$ and $y(t) = 0 \forall t \geq T_p$, i.e. the trajectories of the system (1) reaches the origin within the prescribed-time T_p and stayed there afterward.

This completes the proof of Theorem 4. $\square$

Remark 7. The basic idea behind the construction of the STM (17) is the ETM

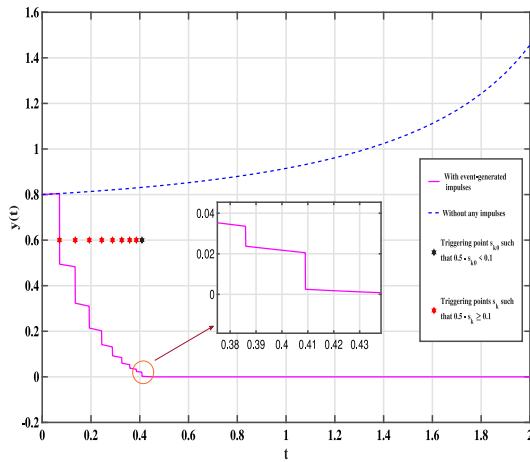
$$s_k = \inf \left\{ t \geq s_{k-1} : \psi(t) \geq \frac{M}{\epsilon} \psi(s_{k-1}) \right\}, \quad (25)$$

and the Eq. (11), obtained from the condition (A2) of Theorem 1. Further, the ETM (25) runs as long as $s_k \leq \mu T_p$. Moreover, in the design of the STM (17) the parameter $L_0 > 1$ is added in order to make sure the existence of at least one triggering instant within the time interval $(0, \mu T_p]$ as mentioned in Remark 6. Further, the contribution of the parameter L_0 is essential as without it, we may not obtain the desired result (i.e., $\psi(\mu T_p) \leq \frac{\epsilon}{M}$).

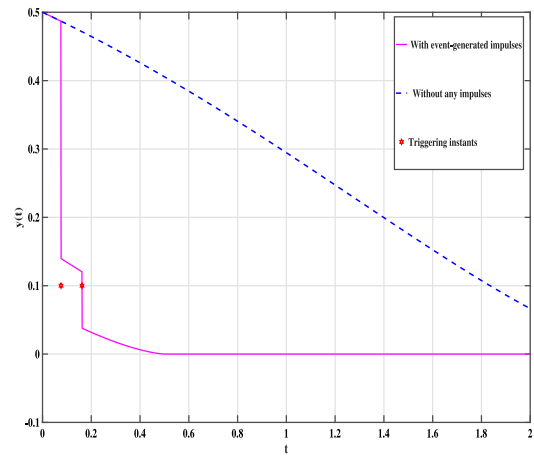
Remark 8. The ETM (2) along with (3) is different from the ETM (25) in the sense that in the first ETM, there may not be any triggering instance and still we can have $\psi(\mu T_p) \leq \frac{\epsilon}{M}$. But, for the ETM (25) this behavior may not be possible as, $\frac{M}{\epsilon} \psi(s_{k-1})$ may not be $\leq \frac{\epsilon}{M}$ at $t = \mu T_p$. To overcome this, an extra parameter L_0 is added in the proposed STM to make sure of occurrence of at least one impulse instant in the interval $(0, \mu T_p]$ and thus the impulse controller gets activated to ensure $\psi(\mu T_p) \leq \frac{\epsilon}{M}$.

Remark 9. It should be noted that the ETM defined in Eq. (2) needs continuous monitoring of the states of the system (1). Thus, it needs separate hardware for this purpose only, which involves the price of excessive cost. Also, it may be difficult to establish a relation between the current triggered instant and the next triggering moment. To overcome these drawbacks, some results on self-triggered mechanisms are discussed in this part of the article. From the STM (17), it can be observed that the next triggering instant can be evaluated based on the information of the current triggered instant. Also it is to be mentioned that in most of the articles on impulsive control theory, the impulsive strengths used to be chosen arbitrarily without any dependency on the other parameters. Similarly, most of the impulsive strengths c_k are chosen arbitrarily for the proposed ETM. But, thinking practically, it would be more meaningful if the impulsive strengths depends on the other parameters. And this is precisely the case for the STM proposed in this article. So, the proposed STM not only ceases the continuous monitoring of the states for the next triggering instant but also have provided an explicit formula to determine the impulsive strengths depending on the other parametric values.

Remark 10. The authors of the article [47] have established a self-triggered strategy, based on a special kind of function, to obtain global asymptotic stability of impulsive systems. Whereas, in this subsection of the paper, we have proposed an STM, based on κ^1 functions, to achieve the prescribed-time stability of the impulsive systems.



(a) Sub-case 1



(b) Sub-case 2

Fig. 2. Simulations of Example 1 corresponding to Theorem 1.

4. Numerical simulations

In this section of the article, two examples are provided to validate the veracity of the results of the present study obtained in the previous sections.

Example 1. The truthfulness of the results on the proposed event-triggered mechanism will be verified through this example. For that, let us consider a ball-motion model with external disturbances and sudden changes in the velocity as follows:

$$\begin{cases} \dot{y}(t) = -\frac{1}{2} |y(t)|^{\frac{1}{3}} (1 + y^2(t))^{\frac{4}{3}} \operatorname{sgn}(y(t)) + |y(t)|^{\frac{5}{3}} (1 + y^2(t))^{\frac{2}{3}} \operatorname{sgn}(y(t)), & t \neq s_k \\ y(t) = \left\{ \frac{c_k y^2(t^-)}{1 + (1 - c_k) y^2(t^-)} \right\}^{\frac{1}{2}} \operatorname{sgn}(y(t^-)), & t = s_k \end{cases} \quad (26)$$

where $y(t)$ is the velocity of the ball and the term $|y(t)|^{\frac{5}{3}} (1 + y^2(t))^{\frac{2}{3}} \operatorname{sgn}(y(t))$ represents the external disturbance that arises from friction. Considering the Lyapunov function as $V = y^2$ and the κ^1 function ψ as $\psi = \frac{V}{1+V}$, one can obtain that the system (26) satisfies the condition (A1) and (A2) of Theorem 1 with $a = 1, b = 2, p = \frac{1}{3}$. Moreover, the impulsive strength c_k ($0 < c_k < 1$) will be defined later on. Now considering $K = 2$ we get, $\epsilon = \frac{1}{8}$. And thus, the stable region is, $|y(t)| \leq \sqrt{\frac{1}{7}} \approx 0.37$. Further we have $A = a \left\{ \frac{K-1}{K} \right\} = \frac{1}{2}$. Now, if one considers $L = 2, \mu_1 = \frac{3}{4}$ and the prescribed-time as $T_p = 2$, then $(A\rho\mu_1 T_p)^{\frac{1}{p}} = \frac{1}{64} < \epsilon = \frac{1}{8}$. Thus we obtain, $M = \frac{\epsilon}{(A\rho\mu_1 T_p)^{\frac{1}{p}}} = 8$. Also, $\mu = 1 - \mu_1 = \frac{1}{4}$ and consider $\epsilon_0 = 0.1 \leq 0.5 = \mu T_p$. Now, according to Theorem 1 consider

$c_k = \frac{1}{2}$ as long as $\mu T_p - s_k = 0.5 - s_k \geq \epsilon_0 = 0.1$ and $c_k = \frac{(\frac{\epsilon}{M})}{e^{b(s_k - s_{k-1})}} = \frac{\frac{1}{64}}{e^{b(s_k - s_{k-1})}}$ when $\{0.5 - s_{k-1} \geq 0.1 \& 0.5 - s_k < 0.1\}$. From Fig. 2(a), one can obtain that, corresponding to the initial condition $y_0 = 0.8$, the trajectory of the system (26) is unstable without any impulse. Whereas, upon applying the ETM (2) along with (3), some impulsive instants are generated, and the trajectory is stabilized within the prescribed-time $T_p = 2$. Further, one can also notice that ETM is stopped when there exists an event-triggered instant s_{k_0} such that $\{0.5 - s_{k_0-1} \geq 0.1 \& 0.5 - s_{k_0} < 0.1\}$. These suffice the existence and validity of sub-case 1 of Theorem 1. Moreover, consider $c_k = 0.1$, keeping all other parametric values intact. Then, one can notice from Fig. 2(b) that for all event-generated impulse instants, we have $\{0.5 - s_k \geq 0.1\}$ corresponding to the initial condition $y_0 = 0.5$, and the trajectory reached the origin within the prescribed-time $T_p = 2$. Whereas, when there was no ETM (2), the trajectory was unable to reach the origin within the provided prescribed-time $T_p = 2$. This justifies the existence and veracity of sub-case2 of Theorem 1.

Example 2. In this example, the veracity of the proposed self-triggered mechanism will be shown. For this, consider Example 1

with $c_k = c = e^{\lfloor \frac{\ln(\frac{M}{\epsilon})}{L_0 b \mu T_p} \rfloor \left\{ \ln\left(\frac{\epsilon}{M}\right) - b \mu T_p \right\}}$ for all k where $L_0 > 1$ and all other parameters are same as defined in Example 1. For numerical simulation, let us consider $L_0 = 5$ (which satisfies the condition $\frac{1}{L_0 b} \ln\left(\frac{M}{\epsilon}\right) < \mu T_p$). Then, we have $s_k - s_{k-1} = \frac{1}{L_0 b} \ln\left(\frac{M}{\epsilon}\right) \approx 0.41$ (by Theorem 3). Hence, according to STM (17) with (18), there is going to be only one self-triggered impulsive instant to make the trajectory of system (26) stabilized within the prescribed-time $T_p = 2$. Further, let us consider $c = 0.01 \leq e^{\lfloor \frac{\ln(\frac{M}{\epsilon})}{L_0 b \mu T_p} \rfloor \left\{ \ln\left(\frac{\epsilon}{M}\right) - b \mu T_p \right\}}$

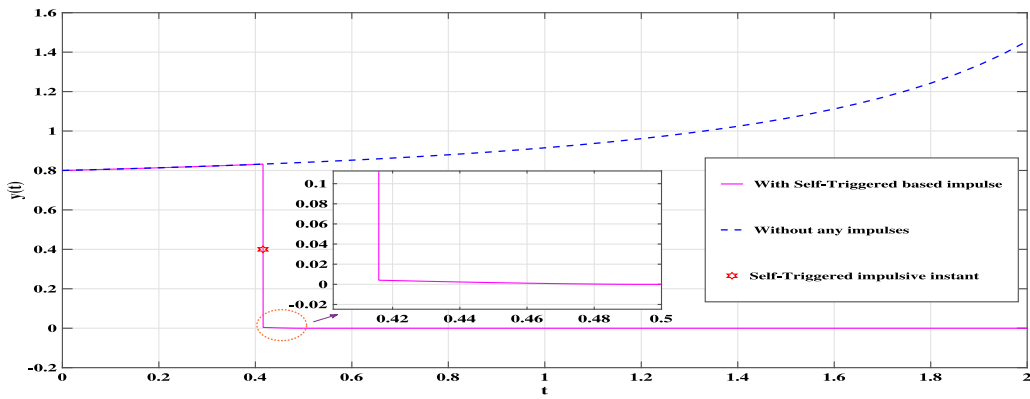
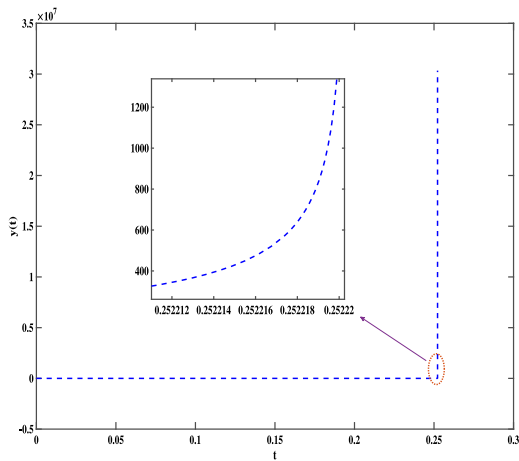
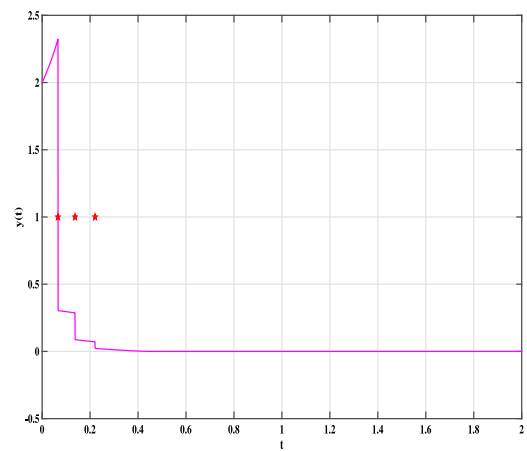


Fig. 3. Simulations of Example 2 with $y_0 = 0.8$ corresponding to Theorem 4.



(a) Example 1 without impulses and with initial condition $y_0 = 2$



(b) Example 1, under the proposed ETM, with initial condition $y_0 = 2$

Fig. 4. Simulations of Remark 11.

for numerical simulation. From Fig. 3, it is clear that without any impulse, the system (26) is unstable. Further, when STM, as mentioned in Section 3.2, is applied, the trajectory of the system (26) is stabilized within the prescribed-time $T_p = 2$. It is also clear from Fig. 3 that only one impulse instant has occurred (which was the theoretical prediction). So, all the theoretical claims of Section 3.2 are verified numerically through Fig. 3.

Remark 11. From Fig. 4(a), one can note that the continuous system of Eq. (26) can show blow-up behavior. For example, if one considers $y_0 = 2$, the solution exists only in the interval $[0, T_a)$ with $T_a \approx 0.25222$. But, upon implementing impulsive instants, generated by the ETM (2) with (3), the impulsive system (26) not only exhibits solution globally on the interval $[0, \infty)$ but also is prescribed-time stable with prescribed-time $T_p = 2$. This fact is verified numerically through Fig. 4(b). Further, similar observations can be obtained in the case of the proposed STM.

5. Conclusion

Prescribed-time stability of impulsive systems via both event-triggered and self-triggered impulsive controls is studied in this article. A Lyapunov-like sufficient condition is presented for the event-triggered mechanism with an exponential decay function. The event-triggered mechanism is designed in such a way that one may obtain desired results in the absence of any triggering instants, also. In the meantime, a self-triggered impulsive control mechanism is also provided to stabilize the impulsive systems within the prescribed time, based on κ^1 functions. A positive minimum inter-event time is present in both processes, ruling out the possibility of Zeno behavior.

Additionally, two instances are given for the numerical verification of theoretical results. In the future, we aim to achieve comparable outcomes for the delayed impulsive systems as well as for the systems subjected to external disturbances. In addition, it will be fascinating to work on adaptive or dynamic triggering in our future studies in the framework of [49].

CRediT authorship contribution statement

Arnab Mapui: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Santwana Mukhopadhyay:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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